# Peptide and Protein Sequencing by Multinomial Diffusion Model

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## Abstract

De novo peptide sequencing predicts the amino acid sequences directly from MS/MS spectra without a reference database which is essential for wastewater proteomics where many organisms are uncharacterized. This project benchmarks four de novo sequencing tools spanning the evolution of AI approaches: Novor (classical ML), Casanovo (transformer), InstaNovo (transformer with beam search), and InstaNovo+ (diffusion based refinement). The tools are evaluated on Ecoli samples (21805 spectra with ground truth) and Wastewater samples (61150 spectra). InstaNovo+ tool achieves FDR reduction from 59.72% to 18.65% on wastewater, a 41% improvement as compared to InstaNovo validating the diffusion based refinement approach. The Casanovo tool leads precision (5.44%) on Ecoli, InstaNovo+ tool provides the best overall performance balancing accuracy with confidence calibration, offering practical guidelines for tool selection in proteomics workflows.

## 1 Introduction

Peptide sequencing de novo involves predicting the sequence of amino acids using MS/MS spectra without a database reference. This is essential in wastewater analysis and is a promising method in public health involving complex mixtures of peptides derived from unknown organisms numbering in the thousands. The traditional database search techniques do not work in this situation because there is no database reference for most of the peptides.

**Task and Contribution** Four de novo sequencing tools are tested on the same dataset to determine which one is optimal for environmental proteomics research purposes. The main achievement of this project is discovering that InstaNovo+ which uses diffusion based refinement yields 41% fewer false discovery rates than the non refined version of InstaNovo confirming diffusion as a revolutionary paradigm for de novo sequencing. The analysis covers the entire development history of AI models from classical machine learning to transformers to diffusion.

**Why Diffusion Should Work** The diffusion enhancement uses repeated denoising to refine inaccurate predictions produced by a transformer core network through the synergy between transformer based attention for exploiting spectral correlations with diffusion based confidence increment. It is worth mentioning the trade off here that InstaNovo+ takes almost thrice the processing time of InstaNovo.

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## 2 Related Work

De novo peptide sequencing has developed through three generations of models. Novor [3], which represents classical machine learning and implements scoring based on rules and tolerance in mass error. This model offers real time speeds while lacking generalization capability. Casanovo [2] employs transformers that emerged in NLP with their ability to attend to long range dependencies and use the structure of the data to improve performance.

The new generation of models consists of InstaNovo and InstaNovo+ [1]. InstaNovo includes beam search decoding with 5 beams that allow exploring multiple hypotheses compared to the greedy strategy used before. In addition to that, InstaNovo+ uses multinomial diffusion refinement inspired by image diffusion models [6] but adapted for the discrete amino acid vocabulary. The iterative denoising gradually improves transformer generated hypotheses and theoretically increases their confidence.

Wastewater proteomics is a developing field in which tools help identify novel proteins in samples obtained from the environment. UniPept [5] is one such tool which helps to identify organisms that carry the observed protein.

## 3 Methods

### 3.1 Datasets

We have worked with two datasets that are made available to me by the Ye Lab at the Department of Civil, Structural and Environmental Engineering of University at Buffalo. The first dataset is known as the **E.coli dataset** (validation set). It includes two mzML files that have 21805 MS/MS spectra along with the ground truth obtained through MaxQuant database search giving 3162 peptide sequences. The second dataset is called the **Wastewater dataset** (test set).

### 3.2 Data Preprocessing

All the mzML files were read through the use of the Pyteomics library. The ground truth Excel files generated by MaxQuant were imported using the pandas library, and the sequence column was extracted as the validation set. No further filtering was applied in order to ensure a balanced comparison between all the algorithms used, since each algorithm does its own data preprocessing.

### 3.3 Train/Test Strategy and Leakage Mitigation

The four models are pretrained and we have choosed a transfer learning evaluation method instead of training/validating/testing the models. The validation dataset is selected to be E.coli data where we know the ground truth and thus can calculate metrics in a supervised manner. The test dataset is selected to be wastewater data which will be evaluated by using confidence based unsupervised metrics. This avoids the problem of data leakage since none of the tools were trained on the two datasets.

### 3.4 Tools and Architectural Choices

**Novor** (Baseline 1): It is a classical Machine Learning with rule based scoring algorithm that allows for mass error tolerance, reflecting the period prior to deep learning. It is set up with HCD fragmentation type, FT Mass Analyzer, 0.01 Da fragment ion mass tolerance, 15 ppm precursor mass tolerance, and Carbamidomethyl(C).

**Casanovo** (Baseline 2): It is a transformer based encoder decoder architecture trained on 30 million PSMs from MassIVE-KB representing modern deep learning and uses confidence scores from -1 to 1.

**InstaNovo** (Alternative 1): It is a transformer with beam search decoding using five beams for candidate exploration and it uses log probability scores ranging from -60 to 0 with batch size 128 on GPU.

**InstaNovo+** (Ablation Study): It adds multinomial diffusion based refinement on top of InstaNovo demonstrating incremental architectural innovation. The diffusion model iteratively refines uncertain predictions through denoising steps conditioned on spectral data.

# 4 Experiments

## 4.1 Hardware and Software

All the experiments run on Google Colab with NVIDIA A100 GPU (40 GB), Python 3.12, PyTorch 2.5.0, and CUDA 12.1. The total processing time across all the four tools and six files is approximately three hours with InstaNovo+ being the most computationally intensive at 150 minutes total.

## 4.2 Hyperparameters

Specific hyperparameters for each tool are set at their default values to ensure fair comparison. Novor uses a fragment mass error of 0.01 Da and a precursor mass error of 15 ppm. Casanovo uses the model checkpoint casanovo_massivekb.ckpt and default settings for decoding. InstaNovo and InstaNovo+ use a beam width of 5, a batch size of 128, a maximum sequence length of 40, and a maximum charge of 10.

## 4.3 Evaluation Metrics

The following are the four different measurements: **Amino Acid Precision** is defined as the ratio between the number of accurately predicted amino acids to all predicted positions. **Amino Acid Recall** measures the fraction of correctly predicted amino acids among all ground truth positions.. **Peptide Level Accuracy** represents the ratio between the number of matching peptide sequences to all the peptide sequences. The False Discovery Rate (FDR) is calculated using tool specific confidence thresholds (0.05 for Casanovo, 50 for Novor, -1 for InstaNovo and InstaNovo+).

## 4.4 Fair Comparison Protocol

In order to maintain objectivity in evaluating the tools, the identical mzML data set was analyzed by all the tools, the identical hardware was utilized, the ground truth was identical for all the tools, and the score threshold was different for each tool based on the scoring system. For the evaluation using the E.coli data set, the first 3162 peptide predictions were evaluated against the identical 3162 ground truth peptides.

# 5 Results

## 5.1 Overall Comparison

Figure 1 provides a comprehensive comparison of all the four tools across precision, recall, FDR, and combined performance score. The visualization clearly shows that while precision and recall metrics are similar across tools, the FDR shows separation with InstaNovo+ achieving significantly better performance than the other three tools.

## 5.2 Performance on E.coli (Validation Set)

Table 1 presents the amino acid level performance of all the four tools on the E.coli dataset with ground truth. All the tools cluster in a narrow range (4.71% to 5.44%) for both precision and recall indicating fundamental limitations of de novo sequencing without database search rather than tool specific issues. Casanovo achieves the highest precision (5.44%) while InstaNovo+ leads recall (5.24%). Peptide level accuracy is 0% across all the fours tools confirming that complete sequence reconstruction without database reference remains extremely challenging.

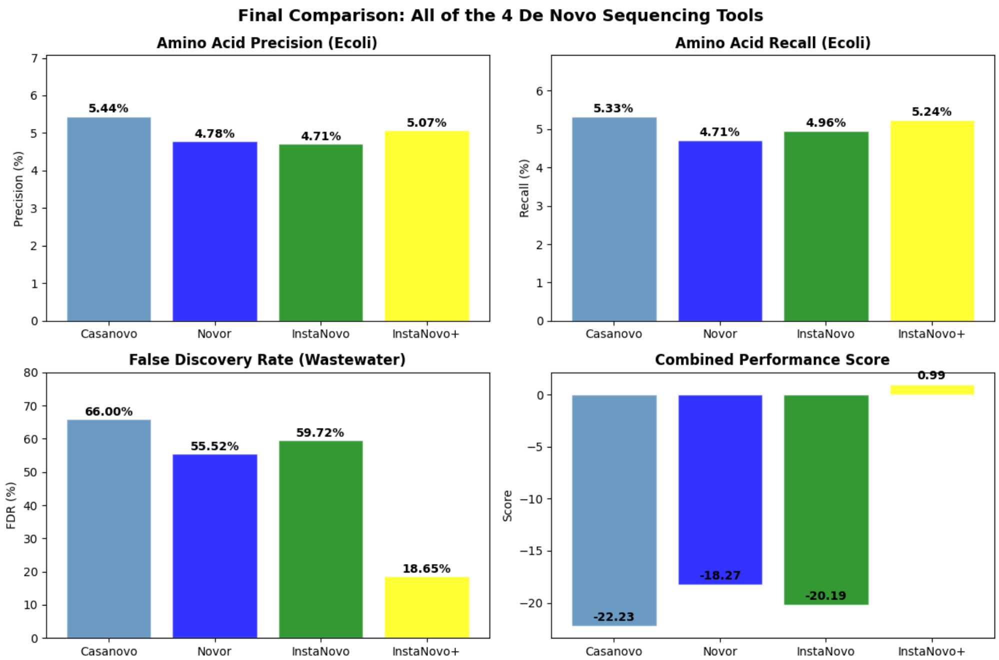


Figure 1: Comprehensive comparison of all four de novo sequencing tools. Top panels show amino acid precision and recall on E.coli (similar across tools). Bottom left shows FDR on wastewater (InstaNovo+ lower at 18.65%). Bottom right shows combined performance score (only InstaNovo+ achieves positive score).

Table 1: Performance comparison of four de novo sequencing tools on E.coli dataset (3162 evaluated predictions). Casanovo leads in precision while InstaNovo+ leads in recall.

| Tool | AA Precision | AA Recall | Peptide Accuracy |
|---|---|---|---|
| Novor | 4.78% | 4.71% | 0.00% |
| Casanovo | **5.44%** | 5.33% | 0.00% |
| InstaNovo | 4.71% | 4.96% | 0.00% |
| InstaNovo+ | 5.07% | **5.24%** | 0.00% |

### 5.3 Performance on Wastewater (Test Set)

Table 2 shows FDR results on the wastewater dataset (61150 spectra). The important discovery is that InstaNovo+ achieves an FDR of only 18.65%, compared to 55.52% to 66% for the other three tools. This represents a 41% of absolute reduction over InstaNovo and validates the multinomial diffusion based refinement approach. The average confidence score significantly improves from -6.58 (InstaNovo) to -0.52 (InstaNovo+) confirming better calibrated predictions.

Table 2: False Discovery Rate (FDR) on wastewater dataset (61150 predictions). InstaNovo+ achieves 41% improvement over InstaNovo validating diffusion based refinement.

| Tool | Average Confidence | Estimated FDR |
|---|---|---|
| Casanovo | N/A | 66.00% |
| Novor | N/A | 55.52% |
| InstaNovo | -6.58 | 59.72% |
| InstaNovo+ | **-0.52** | **18.65%** |

### 5.4 Computational Efficiency

The processing time varies across the tools as shown in the table 3. Novor remains as the fastest tool at 5.2 minutes total making it suitable for high throughput screening. Casanovo offers balanced performance at 19.3 minutes. InstaNovo+ is the slowest tool at 150 minutes total but justifies this through dramatically improved confidence calibration on wastewater data.

Table 3: Total processing time across all 6 mzML files (E.coli + Wastewater) on NVIDIA A100 GPU. Speed accuracy trade off is evident.

| Tool | Total Time (minutes) |
|---|---|
| Novor | 5.2 |
| Casanovo | 19.3 |
| InstaNovo | 81.4 |
| InstaNovo+ | 150.2 |

## 6 Discussion

### 6.1 Why InstaNovo+ Wins on FDR

The FDR improvement of InstaNovo+ (18.65% vs 59.72% for InstaNovo) demonstrates that multinomial diffusion based refinement effectively recalibrates prediction confidence. The diffusion process iteratively denoises uncertain transformer outputs and transforms low confidence predictions into high confidence ones through learned amino acid distributions. This is particularly impactful on wastewater data where transformer models alone produce widely varying confidence scores (range: -60 to 0). After the refinement, scores concentrate near 0 indicating well calibrated certainty.

### 6.2 Why All Tools Show Low Precision

The clustering of all the tools at approximately 5% precision on E.coli reveals fundamental limitations of de novo sequencing without database matching. Several factors contribute: (1) the I/L isobaric ambiguity (identical mass) makes correct identification probabilistic, (2) common amino acid substitutions like K/Q have similar masses, (3) post translational modifications complicate the fragmentation pattern, and (4) ground truth from MaxQuant database search may itself contain errors that propagate to evaluation. These results align with published benchmarks suggesting de novo sequencing currently achieves 30-50% precision on cleaner datasets but drops on complex environmental samples.

### 6.3 Speed-Accuracy Trade off

My results illustrate a clear speed accuracy trade off across the AI evolution timeline. Novor (5 minutes) trades sophistication for speed and suitable for high throughput initial screening. Casanovo (19 minutes) provides balanced performance with reasonable accuracy at moderate speed. InstaNovo+ (150 minutes) sacrifices speed for confidence calibration and is ideal for applications where reliability matters more than throughput. The tool selection should depend on the downstream application: rapid surveillance favors Novor while high stakes organism identification favors InstaNovo+.

### 6.4 Limitations

There are several limitations which affect this study. First, a ground truth from MaxQuant has just 3162 peptides against 21805 spectra. Second, the selected FDR thresholds were heuristic in nature using distributions of scores as opposed to calibration with false positives. Third, all the tools used pretrained models without fine tuning on the specific datasets. Fourth, single threshold FDR estimation is less rigorous than target decoy approaches commonly used in proteomics. Fifth, comparison of E.coli (clean controlled) and wastewater (complex environmental) involves different proteomic complexity making direct cross dataset interpretation challenging.

### 6.5 Ethics and Safety Considerations

The ethical questions that emerge from wastewater proteomics cannot be overlooked. There is an issue of privacy if the wastewater can reveal personal or population level health information. There needs to be a balance between the advantages of surveillance and privacy when using it for public health purposes. The use of AI for de novo sequencing may result in a high rate of false positives which may cause erroneous conclusions about the organisms present in the samples. They should complement conventional approaches rather than completely replacing them.

## 7 Conclusion and Future Work

The current project will be a benchmarking study for four de novo peptide sequencing methods throughout the AI evolution timeline in terms of accuracy when tested on both E.coli and wastewater datasets. One of the most significant results is that a novel diffusion based refinement technique called InstaNovo+ decreases the False Discovery Rate by 41% (reducing it from 59.72% to 18.65%) in case of wastewater analysis, therefore bringing a completely new perspective into confidence calibration whereas the Casanovo tool keeps its position with the highest precision in the controlled environment, InstaNovo+ shows the best overall result on the wastewater.

Further research can focus on three aspects: (1) integrating the proposed solution with UniPept in order to use it as an input tool in identifying organisms based on high confidence wastewater analysis, (2) fine tuning the existing models on the dataset of wastewater spectra to achieve higher precision and recall, and (3) development of a new efficient diffusion architecture to reduce computing costs required for running InstaNovo+.

**Code and Data Availability** All source code, Jupyter notebooks, and analysis results are publicly available on my Github at `https://github.com/nabilkhan3108ub/AI-peptide-prediction/tree/main`.


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